

WINDOWS 11 FIREWALL & NETWORK PROTECTION

Hands-On Security Lab — Portfolio Project

SOC Analyst Training Series

Platform: Windows 11 Pro | Date: March 21–22, 2026

Tools: wf.msc · PowerShell · Windows Security · Event Viewer

README — Project Overview

This project documents a complete hands-on lab exercise covering Windows 11 firewall configuration and network protection. The lab was designed to build practical experience with the exact scenarios encountered by SOC analysts and system administrators when investigating reports of blocked or inaccessible websites.

Objectives

- Understand the difference between Windows Defender Firewall outbound rules and Windows Defender Network Protection (SmartScreen)
- Create and manage firewall rules using both PowerShell and the wf.msc GUI
- Observe and document the browser error messages produced by each type of block
- Read and interpret firewall drop logs (pfirewall.log) and Event Viewer audit trails
- Remediate firewall blocks and restore connectivity
- Enable, test, and disable Network Protection (SmartScreen) in both Audit and Enforce modes

Lab Environment

Component	Details
Platform	Windows 11 Pro (Dell hardware)
Browser	Microsoft Edge (Chromium)
Shell	Windows PowerShell (Administrator)
Firewall Console	wf.msc — Windows Defender Firewall with Advanced Security
Security Console	Windows Security (windowsdefender://)
Test Target	https://example.com (IANA reserved domain)
NP Test URL	https://smartscreentestratings2.net (Microsoft official test domain)
Date Performed	21–22 March 2026

Skills Demonstrated

- DNS resolution with PowerShell (Resolve-DnsName)
- Firewall outbound rule creation — PowerShell (New-NetFirewallRule) and GUI wizard
- Firewall drop log analysis (pfirewall.log)
- Event Viewer firewall audit trail reading (Event ID 2082)
- Rule verification and remediation (Remove-NetFirewallRule / GUI Delete)
- Network Protection state management (Set-MpPreference)
- SmartScreen behaviour observation and toggle via Windows Security GUI

Project Files

File	Description
<code>README.md</code>	Project overview, objectives, environment (this section)
<code>commands.md</code>	All PowerShell and CLI commands used in sequence
<code>lab-report.docx</code>	Full illustrated lab report with screenshots (this document)

Command Reference — All Commands in Sequence

All commands were run in Windows PowerShell (Administrator). Commands are presented in execution order with expected output noted.

Phase 1 — Baseline & DNS Resolution

Step 1.1 — Confirm target site loads

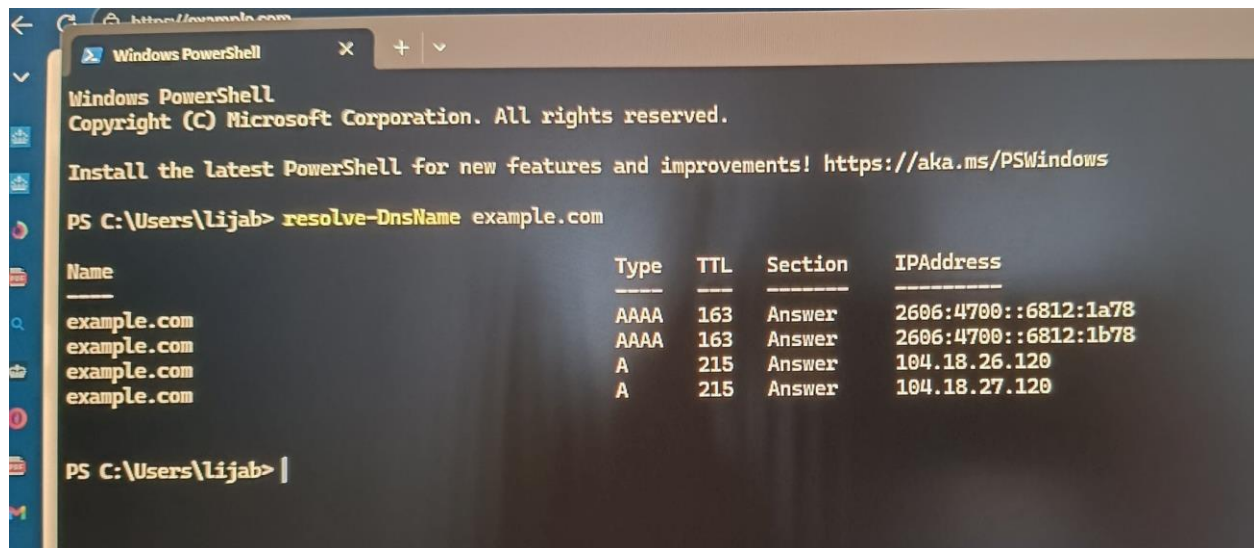
Navigate to <https://example.com> in Edge. Confirm page loads before making any changes.

Step 1.2 — Resolve DNS for target

```
PS C:\> Resolve-DnsName example.com
```

NOTE

Expected output: 2x AAAA records (IPv6) and 2x A records (IPv4). Note all 4 IPs — you must block all of them or the browser will use an unblocked one.



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\lijab> resolve-DnsName example.com

Name                                     Type  TTL  Section  IPAddress
-----
example.com                             AAAA   163  Answer   2606:4700::6812:1a78
example.com                             AAAA   163  Answer   2606:4700::6812:1b78
example.com                             A      215  Answer   104.18.26.120
example.com                             A      215  Answer   104.18.27.120

PS C:\Users\lijab> |
```

Figure 1 — Resolve-DnsName showing 4 addresses for example.com

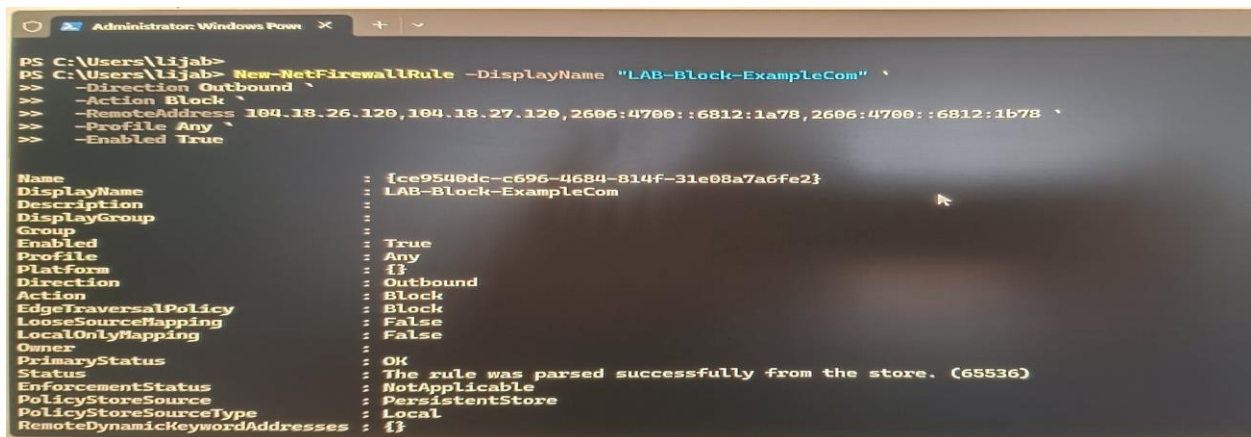
Phase 2 — Block via PowerShell

Step 2.1 — Create outbound block rule

```
PS C:\> New-NetFirewallRule -DisplayName "LAB-Block-ExampleCom" `
  -Direction Outbound `
  -Action Block `
  -RemoteAddress 104.18.26.120,104.18.27.120,2606:4700::6812:1a78,2606:4700::6812:1b78 `
  -Profile Any `
  -Enabled True
```

i INFO

No output confirmation prompt. Rule is immediately active. All 4 IPs (both IPv4 and IPv6) must be included in a single rule.



```
PS C:\Users\liljab> New-NetFirewallRule -DisplayName "LAB-Block-ExampleCom" `
>> -Direction Outbound `
>> -Action Block `
>> -RemoteAddress 104.18.26.120,104.18.27.120,2606:4700::6812:1a78,2606:4700::6812:1b78 `
>> -Profile Any `
>> -Enabled True

Name : {ce9540dc-c696-4684-814f-31e08a7a6fe2}
DisplayName : LAB-Block-ExampleCom
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : {}
Direction : Outbound
Action : Block
EdgeTraversalPolicy :
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses : {}
```

Figure 2 — New-NetFirewallRule executed successfully, PrimaryStatus: OK

Step 2.2 — Test connectivity (confirm block is active)

```
PS C:\> Test-NetConnection -ComputerName 104.18.26.120 -Port 443
```

✓ RESULT

Expected: TcpTestSucceeded : False — confirms TCP connection is being dropped by the firewall.

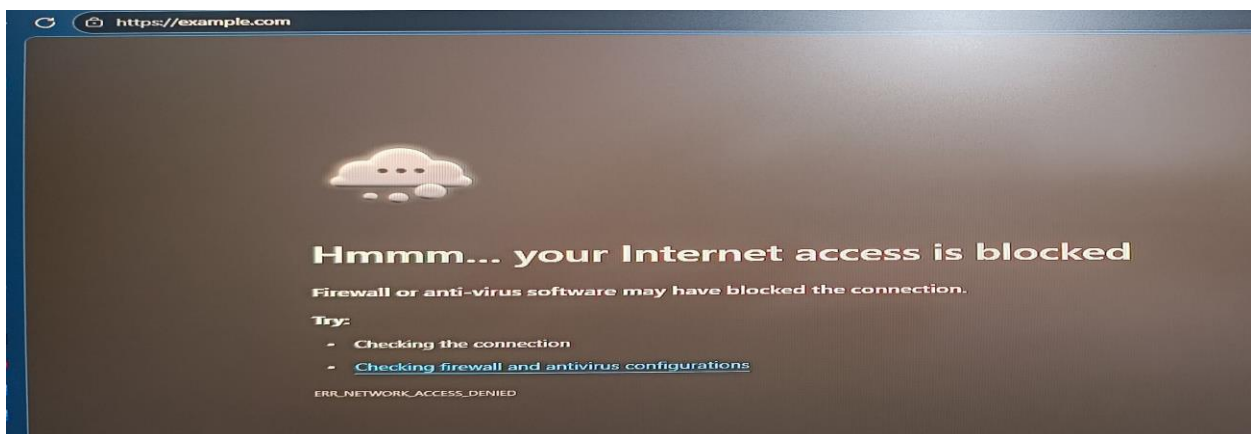


Figure 3 — Browser showing ERR_NETWORK_ACCESS_DENIED after rule creation

Phase 3 — Enable Firewall Drop Logging

Step 3.1 — Enable logging via netsh

```
PS C:\> netsh advfirewall set allprofiles logging droppedconnections enable
```

Step 3.2 — Verify logging is active

```
PS C:\> netsh advfirewall show allprofiles | findstr -i "log"
```

i INFO

Expected: LogDroppedConnections: Enable for all 3 profiles (Domain, Private, Public). Log path: %systemroot%\system32\LogFiles\Firewall\pfirewall.log

```
PS C:\Users\lijab> netsh advfirewall set allprofiles logging droppedconnections enable
Ok.

PS C:\Users\lijab> netsh advfirewall show allprofiles | findstr -i "log"
Logging:
LogAllowedConnections      Disable
LogDroppedConnections      Enable
FileName                   %systemroot%\system32\LogFiles\Firewall\pfirewall.log
Logging:
LogAllowedConnections      Disable
LogDroppedConnections      Enable
FileName                   %systemroot%\system32\LogFiles\Firewall\pfirewall.log
Logging:
LogAllowedConnections      Disable
LogDroppedConnections      Enable
FileName                   %systemroot%\system32\LogFiles\Firewall\pfirewall.log
PS C:\Users\lijab>
```

Figure 4 — Logging enabled across all profiles via netsh

Step 3.3 — Read the drop log

```
PS C:\> Get-Content "$env:SystemRoot\system32\LogFiles\Firewall\pfirewall.log" | Select-Object -Last 20
```

```
Windows PowerShell
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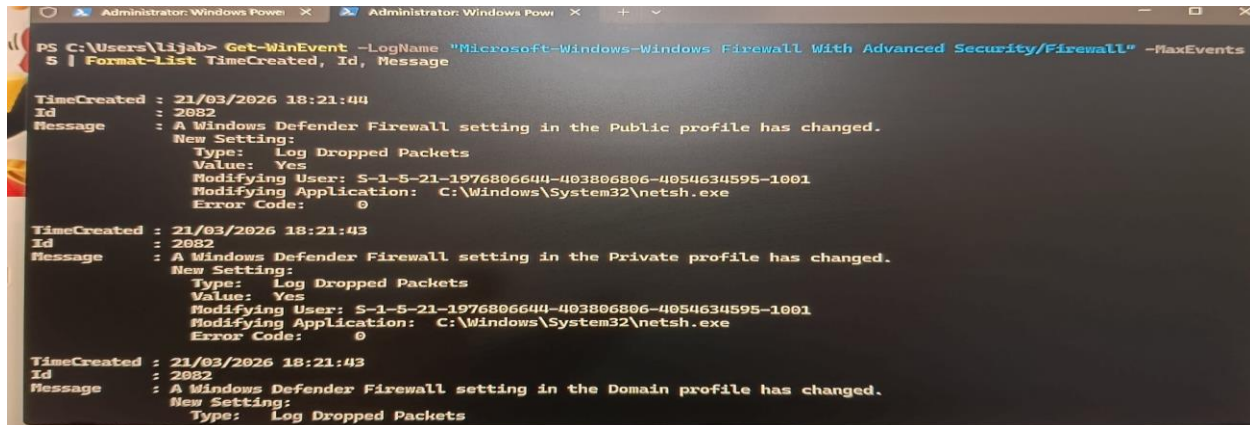
PS C:\Users\lijab> Get-Content "$env:SystemRoot\system32\LogFiles\Firewall\pfirewall.log" | Select-Object -Last 20
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1b78 55302 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1a78 54846 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.26.120 50036 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.27.120 62055 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1b78 49971 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1a78 55550 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.26.120 49491 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.27.120 57820 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1b78 59664 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1a78 50889 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.26.120 59472 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.27.120 60710 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1b78 52724 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 2a00:23c8:4429:9401:f041:7b3e:d37e:734c 2606:4700::6812:1a78 51632 443 0 - 0 0 0 - - SEND
904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.26.120 60735 443 0 - 0 0 0 - - SEND 904
2026-03-21 18:11:54 DROP TCP 192.168.1.65 104.18.27.120 56591 443 0 - 0 0 0 - - SEND 904
```

Figure 5 — pfirewall.log showing DROP entries for all 4 IPs on port 443

Phase 4 — Event Viewer Audit Trail

Step 4.1 — Query firewall setting change events

```
PS C:\> Get-WinEvent -LogName "Microsoft-Windows-Windows Firewall With Advanced Security/Firewall" -MaxEvents 5 |  
Format-List TimeCreated, Id, Message
```



```
PS C:\Users\liljab> Get-WinEvent -LogName "Microsoft-Windows-Windows Firewall With Advanced Security/Firewall" -MaxEvents 5 | Format-List TimeCreated, Id, Message  
  
TimeCreated : 21/03/2026 18:21:44  
Id           : 2082  
Message      : A Windows Defender Firewall setting in the Public profile has changed.  
               New Setting:  
                 Type: Log Dropped Packets  
                 Value: Yes  
                 Modifying User: S-1-5-21-1976806644-403806806-4054634595-1001  
                 Modifying Application: C:\Windows\System32\netsh.exe  
                 Error Code: 0  
  
TimeCreated : 21/03/2026 18:21:43  
Id           : 2082  
Message      : A Windows Defender Firewall setting in the Private profile has changed.  
               New Setting:  
                 Type: Log Dropped Packets  
                 Value: Yes  
                 Modifying User: S-1-5-21-1976806644-403806806-4054634595-1001  
                 Modifying Application: C:\Windows\System32\netsh.exe  
                 Error Code: 0  
  
TimeCreated : 21/03/2026 18:21:43  
Id           : 2082  
Message      : A Windows Defender Firewall setting in the Domain profile has changed.  
               New Setting:  
                 Type: Log Dropped Packets
```

Figure 6 — Event ID 2082 entries showing firewall setting changes with timestamps and tool used

⚠ NOTE

Event ID 2082 records every firewall setting change. The log shows both mmc.exe (GUI) and netsh.exe changes with exact timestamps — key forensic evidence in a real investigation.

Phase 5 — Remediation (PowerShell)

Step 5.1 — Remove the block rule

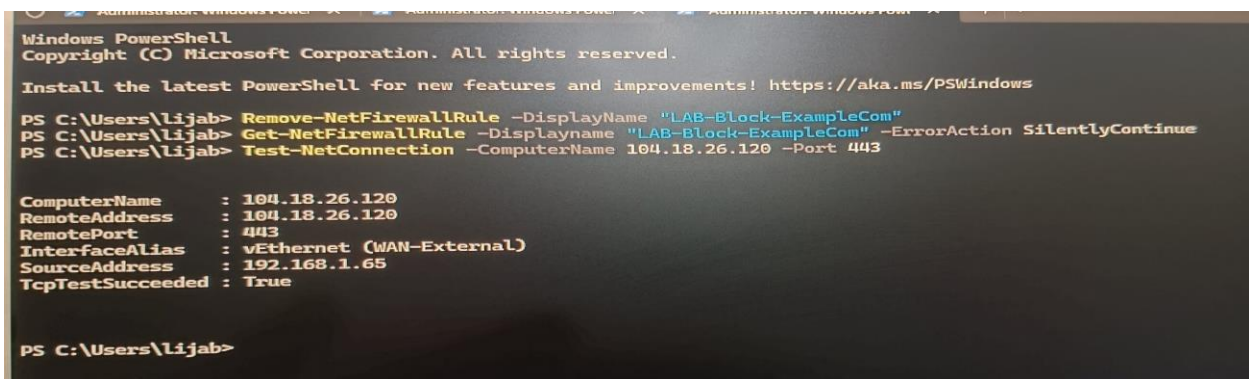
```
PS C:\> Remove-NetFirewallRule -DisplayName "LAB-Block-ExampleCom"
```

Step 5.2 — Verify rule is removed

```
PS C:\> Get-NetFirewallRule -DisplayName "LAB-Block-ExampleCom" -ErrorAction SilentlyContinue
PS C:\> Test-NetConnection -ComputerName 104.18.26.120 -Port 443
```

✓ RESULT

Expected: Get-NetFirewallRule returns nothing (rule gone). TcpTestSucceeded : True confirms connectivity restored.



```
Windows PowerShell
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PS C:\Users\lijab> Remove-NetFirewallRule -DisplayName "LAB-Block-ExampleCom"
PS C:\Users\lijab> Get-NetFirewallRule -Displayname "LAB-Block-ExampleCom" -ErrorAction SilentlyContinue
PS C:\Users\lijab> Test-NetConnection -ComputerName 104.18.26.120 -Port 443

ComputerName           : 104.18.26.120
RemoteAddress          : 104.18.26.120
RemotePort             : 443
InterfaceAlias         : vEthernet (WAN-External)
SourceAddress          : 192.168.1.65
TcpTestSucceeded       : True

PS C:\Users\lijab>
```

Figure 7 — Rule removed, TcpTestSucceeded: True, connectivity confirmed

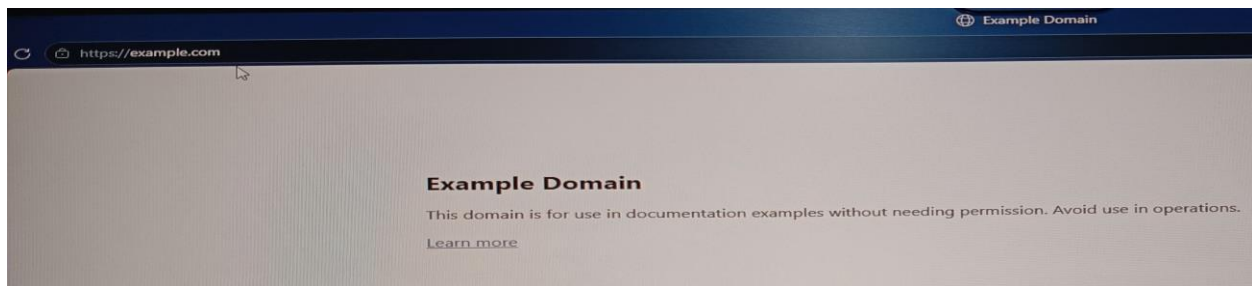


Figure 8 — example.com loads again in Edge after rule removal

Phase 6 — Block via GUI (wf.msc Wizard)

The same block was recreated using the Windows Defender Firewall GUI to build muscle memory for the interface.

Step 6.1 — Open wf.msc

```
Press Win+R → type: wf.msc → Enter
Navigate to: Outbound Rules → right-click → New Rule...
```

Step 6.2 — GUI Wizard sequence

Wizard Page	Selection
Rule Type	Custom
Program	All programs
Protocol & Ports	Any (leave default)
Scope - Local	Any IP address
Scope - Remote	These IP addresses → Add all 4 IPs
Action	Block the connection
Profile	Domain ✓ Private ✓ Public ✓
Name	LAB-Block-ExampleCom-GUI

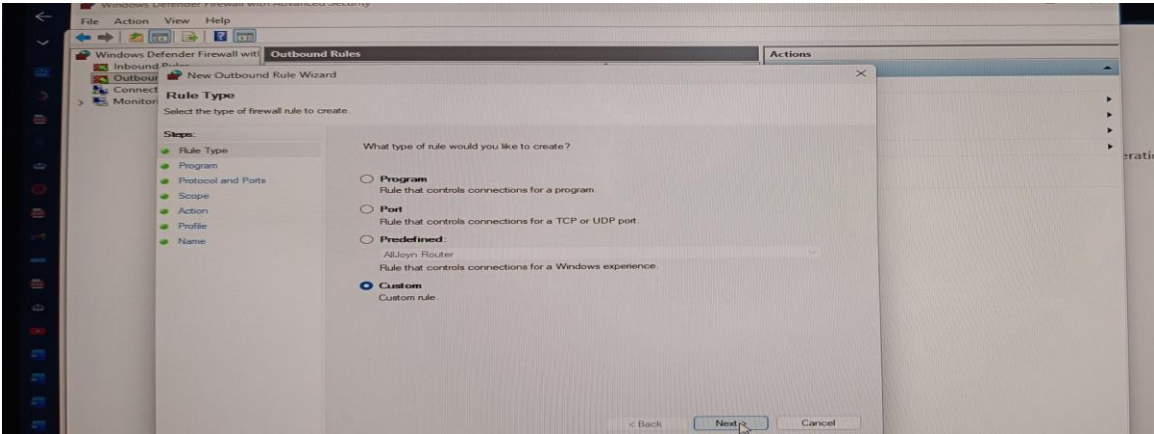


Figure 9 — Rule Type: Custom selected

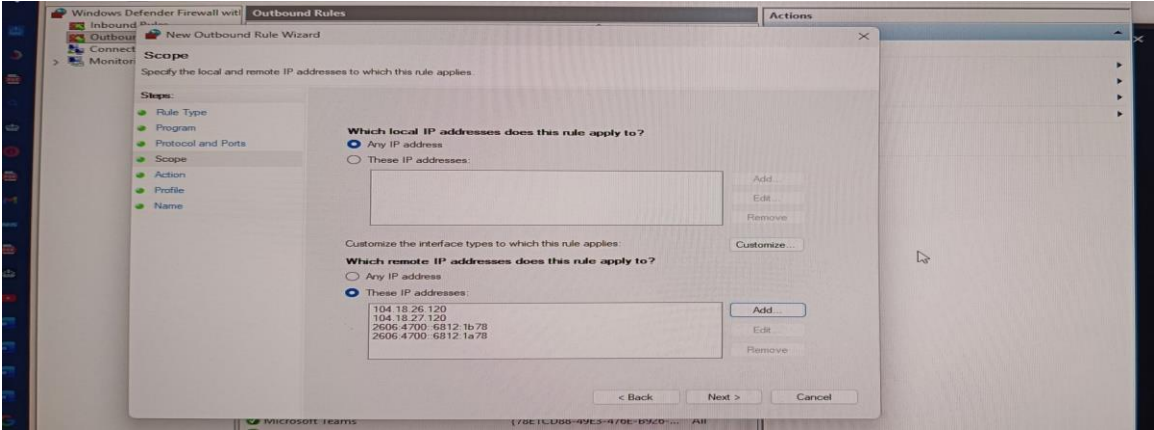


Figure 10 — Scope page: all 4 IPs entered under Remote IP addresses

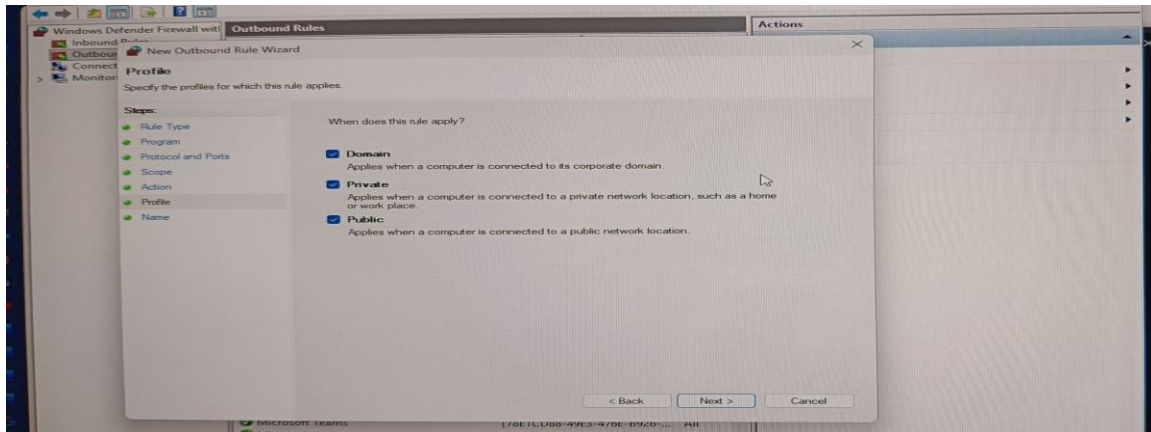


Figure 11 — Profile page: all three profiles checked

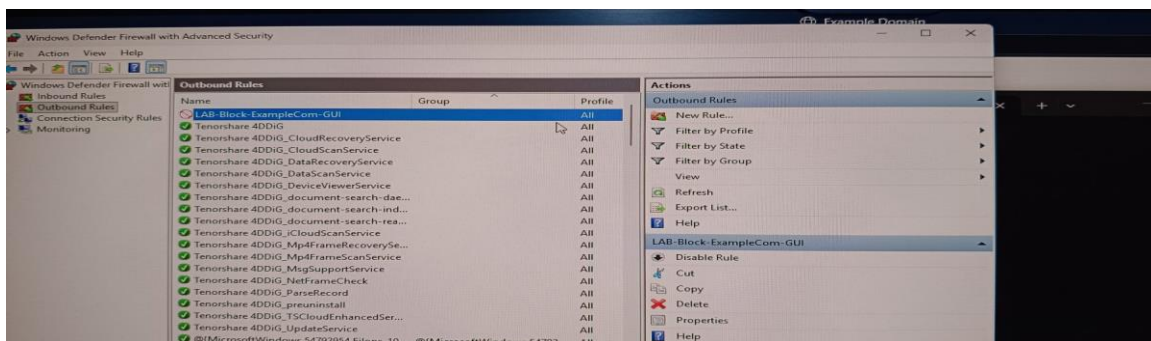


Figure 12 — Rule created, visible in Outbound Rules list with red block icon

Phase 7 — Rule Inspection via GUI

Step 7.1 — Inspect rule Properties

Double-click the rule in wf.msc to open Properties. Review all tabs.

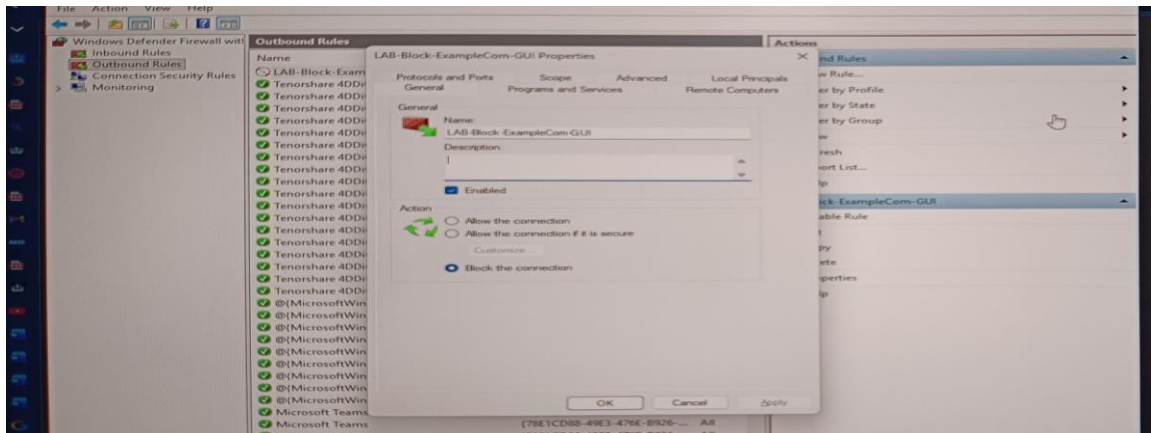


Figure 13 — General tab: Action = Block the connection, Enabled = checked

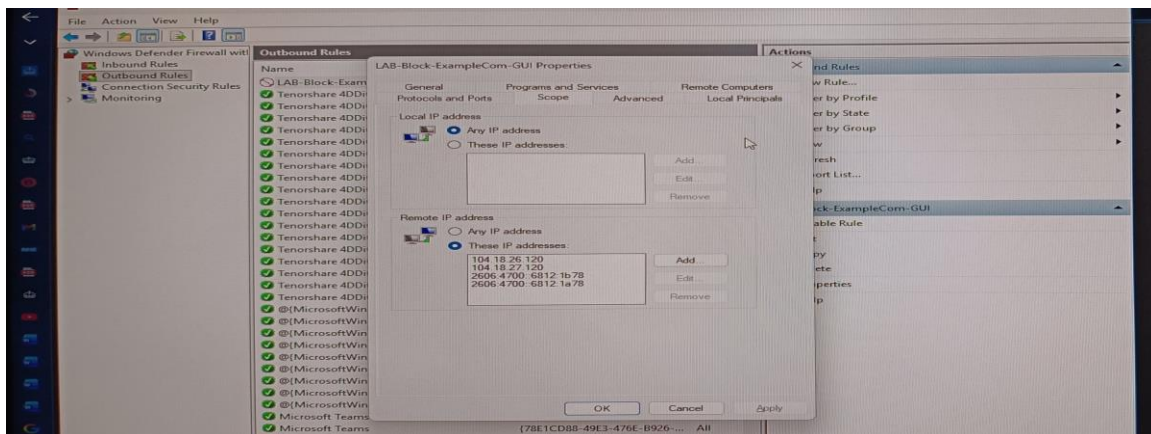


Figure 14 — Scope tab: all 4 IPs listed under Remote IP address

Step 7.2 — Delete rule via GUI

wf.msc → Outbound Rules → right-click LAB-Block-ExampleCom-GUI → Delete → Yes

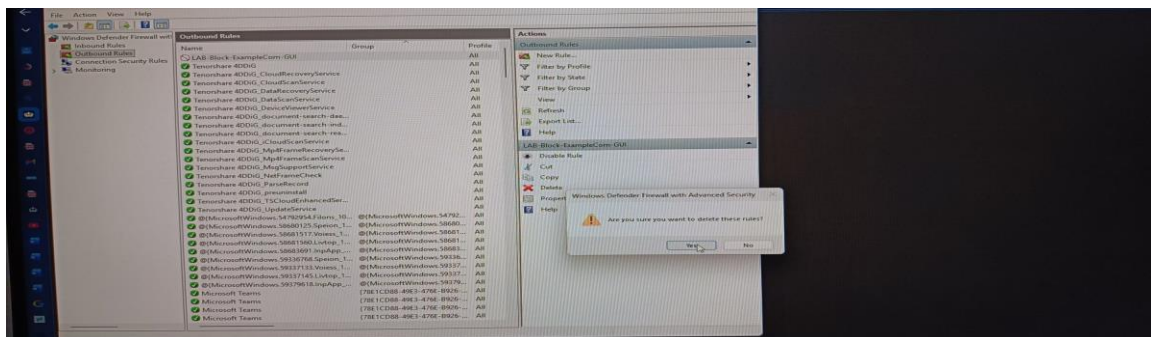


Figure 15 — Confirmation dialog before deleting rule



Phase 8 — Network Protection (SmartScreen)

Step 8.1 — Check current state

```
PS C:\> Get-MpPreference | Select-Object EnableNetworkProtection
# Returns: 0 (Disabled) 1 (Enabled) 2 (Audit Mode)
```

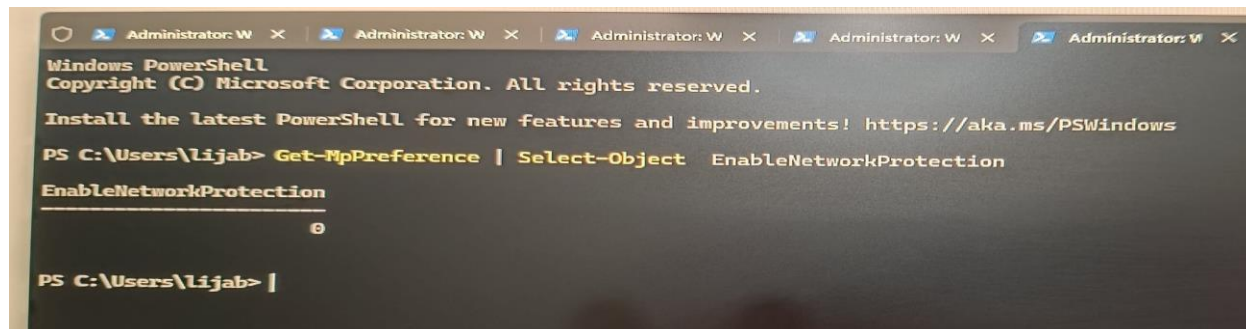


Figure 17 — EnableNetworkProtection: 0 (Disabled by default)

Step 8.2 — Enable Audit Mode (safe for testing)

```
PS C:\> Set-MpPreference -EnableNetworkProtection AuditMode
PS C:\> Get-MpPreference | Select-Object EnableNetworkProtection
# Returns: 2
```

Step 8.3 — Enable full enforcement

```
PS C:\> Set-MpPreference -EnableNetworkProtection Enabled
PS C:\> Get-MpPreference | Select-Object EnableNetworkProtection
# Returns: 1
```

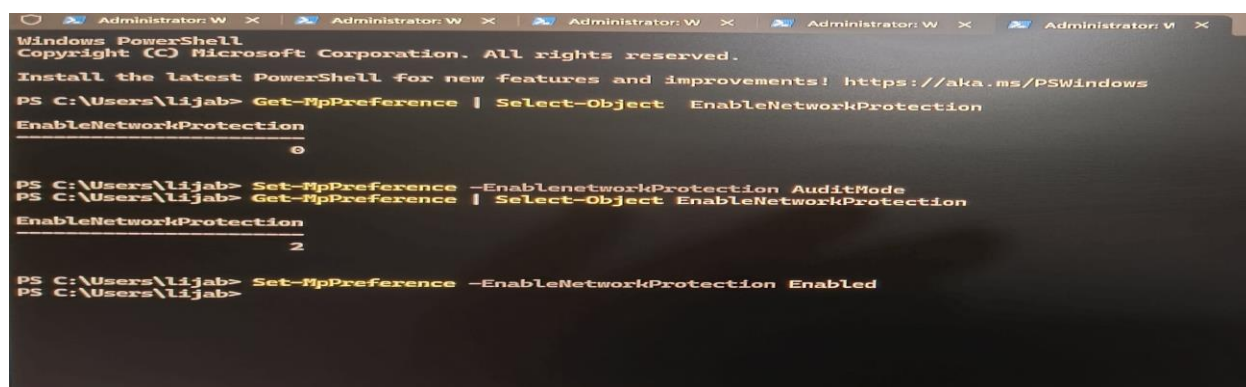


Figure 18 — Progression: 0 → AuditMode (2) → Enabled (1)

Step 8.4 — Test with Microsoft's official NP test URL

```
Navigate to: https://smartscreentestratings2.net
```

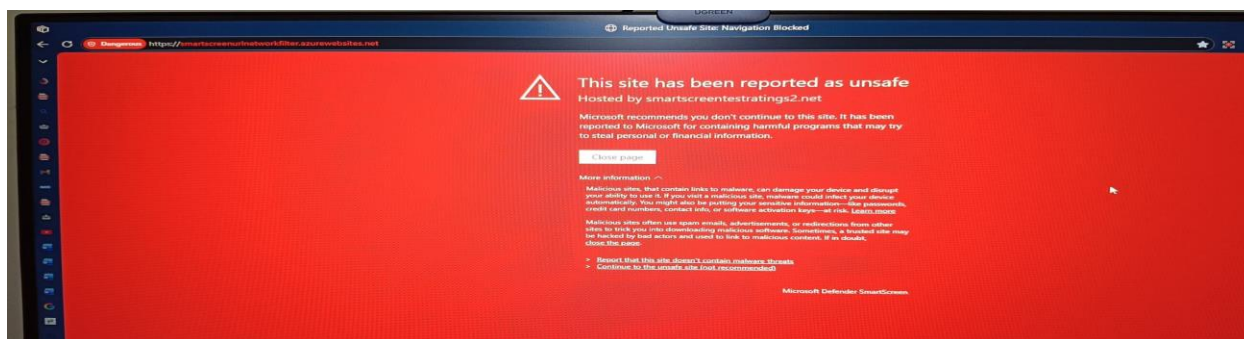


Figure 19 — SmartScreen full red block page: 'This site has been reported as unsafe'

Step 8.5 — Restore original state

```
PS C:\> Set-MpPreference -EnableNetworkProtection Disabled
PS C:\> Get-MpPreference | Select-Object EnableNetworkProtection
# Returns: 0
```

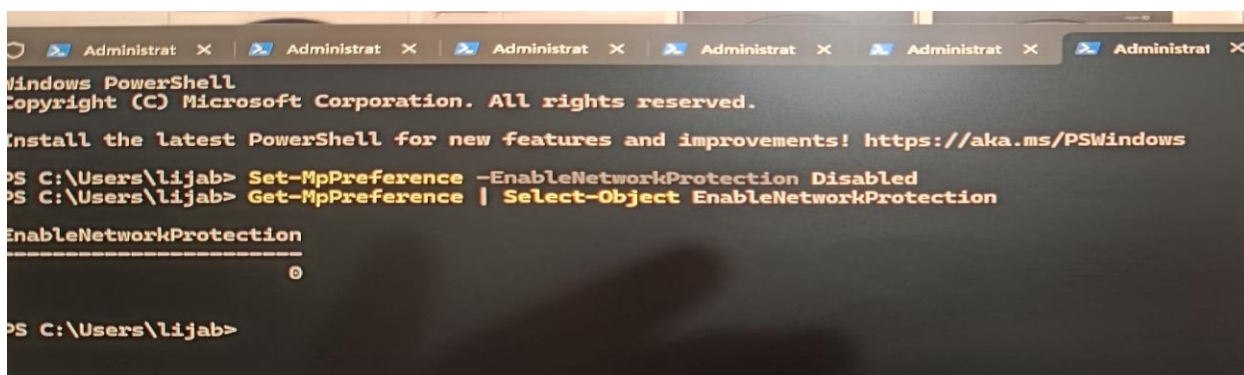


Figure 20 — Network Protection restored to Disabled (0)

Phase 9 — Network Protection via Windows Security GUI

Step 9.1 — Navigate to SmartScreen settings

Windows Security → App & browser control → Reputation-based protection settings

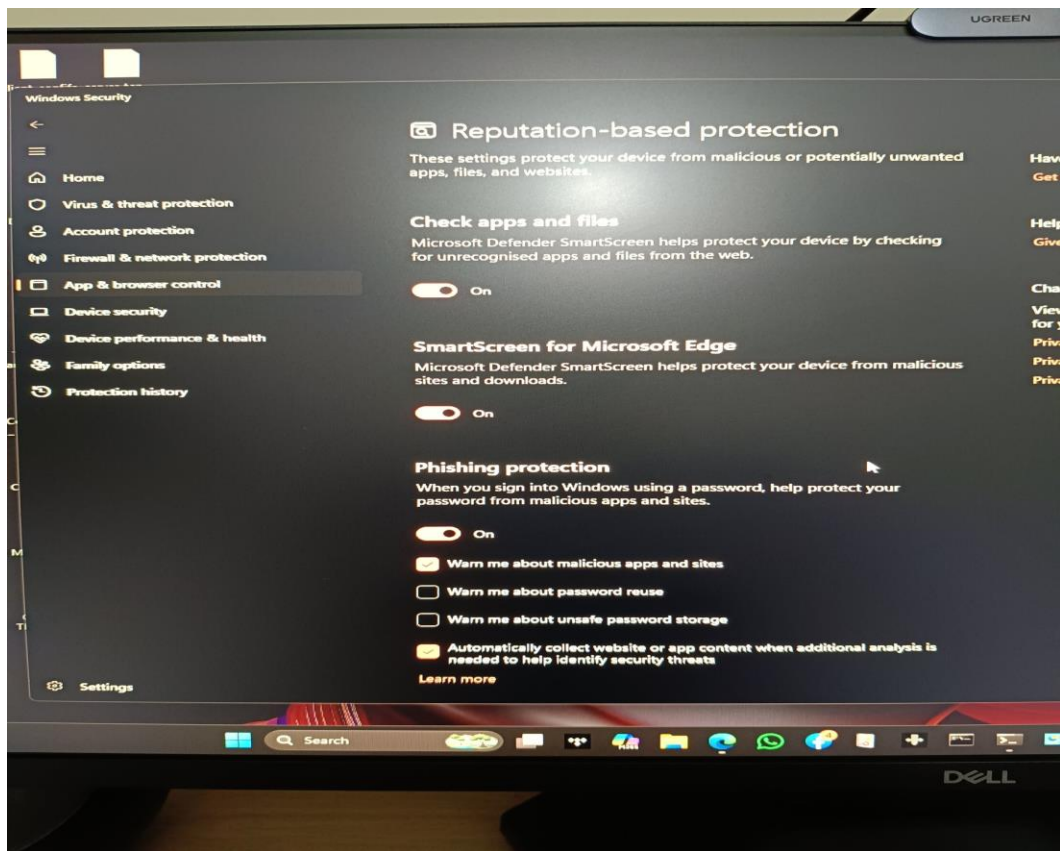


Figure 21 — Reputation-based protection settings: Check apps and files, SmartScreen for Edge, Phishing protection

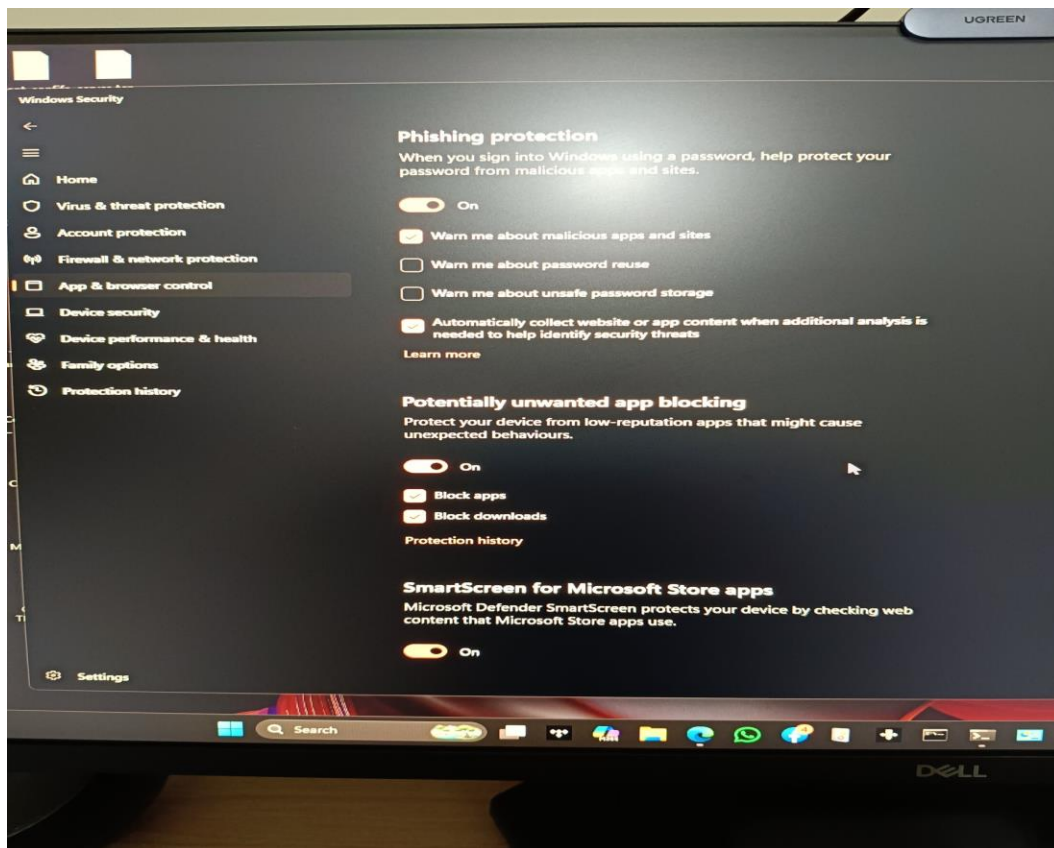


Figure 22 — Scrolled down: Potentially unwanted app blocking, SmartScreen for Microsoft Store apps

i INFO

The 'SmartScreen for Microsoft Edge' toggle is the GUI equivalent of the Network Protection test. Note: the master EnableNetworkProtection toggle is only available via PowerShell or Group Policy in this Windows build.

Analysis & Key Findings

Error Signature Comparison

Understanding which error a user reports is the first triage step. The table below maps errors to their causes:

Browser Error	Cause	Behaviour
<code>ERR_NETWORK_ACCESS_DENIED</code>	Firewall outbound rule block	Immediate — WFP intercepts at socket level before packet leaves machine
<code>ERR_CONNECTION_TIMED_OUT</code>	Firewall block (silent drop)	Slow ~30s — packet sent but silently discarded
Red SmartScreen page	Network Protection / SmartScreen	Instant — reputation check at DNS/URL layer
<code>ERR_NAME_NOT_RESOLVED</code>	DNS block or NXDOMAIN	Instant — domain not resolved at all
<code>ERR_CONNECTION_REFUSED</code>	Active port rejection	Instant — RST packet received

Firewall vs Network Protection — Diagnostic Differences

Property	Firewall Rule vs Network Protection
Block mechanism	IP address (layer 3/4) vs Domain reputation (layer 7/DNS)
Configuration	wf.msc / New-NetFirewallRule vs Set-MpPreference / Windows Security GUI
Block speed	Slow (timeout) or instant (WFP) vs Always instant
DNS resolution	Succeeds (IP known, connection blocked) vs Usually also blocked
Log location	pfirewall.log + Event ID 5157 vs Defender Operational Event 1125/1126
Bypass risk	Browser tries all IPs — must block all vs Domain-level, harder to bypass
Fix	Remove-NetFirewallRule / GUI Delete vs Whitelist URL or disable NP

SOC Analyst Triage Workflow

When a user reports 'website won't open', follow this diagnostic chain:

1. DNS check: `Resolve-DnsName <domain>` — does it resolve to an IP?
2. TCP test: `Test-NetConnection -ComputerName <IP> -Port 443` — does `TcpTestSucceeded` return `True` or `False`?

3. If False: Check pfirewall.log for DROP entries to that IP on port 443
4. If DROP entries found: `Get-NetFirewallRule | Where-Object { $_.Action -eq 'Block' }` to find the blocking rule
5. Identify rule source (who created it, when) via Event ID 2082 in the Firewall event log
6. Remove rule if appropriate: `Remove-NetFirewallRule -DisplayName '<name>'`
7. If browser shows red SmartScreen page instead: check Network Protection state and SmartScreen settings in Windows Security

Observations from Lab

- example.com returned 4 addresses (2 IPv4 + 2 IPv6). Blocking only the IPv4 addresses would have allowed the browser to connect via IPv6 — all addresses must be blocked simultaneously.
- The `ERR_NETWORK_ACCESS_DENIED` error (vs the expected `ERR_CONNECTION_TIMED_OUT`) occurred because Edge attempted IPv6 first, which was blocked at the WFP layer before the TCP handshake could begin — producing an immediate rejection rather than a timeout.
- Event ID 2082 recorded both GUI (`mmc.exe`) and command-line (`netsh.exe`) changes with precise timestamps — demonstrating the value of the firewall audit log for forensic reconstruction.
- Network Protection (`value=0`) was disabled by default on this standalone Windows 11 machine — consistent with Microsoft's default configuration for non-enterprise deployments.
- The GUI SmartScreen toggle (App & browser control → Reputation-based protection) controls the Edge-level block. The master Network Protection setting requires PowerShell or Group Policy.

Appendix — Quick Reference

All Commands at a Glance

```
# — DNS & Connectivity —————
Resolve-DnsName example.com
Test-NetConnection -ComputerName 104.18.26.120 -Port 443

# — Create Firewall Block Rule —————
New-NetFirewallRule -DisplayName "LAB-Block-ExampleCom" `
  -Direction Outbound -Action Block `
  -RemoteAddress 104.18.26.120,104.18.27.120,2606:4700::6812:1a78,2606:4700::6812:1b78 `
  -Profile Any -Enabled True

# — Verify / Remove Rule —————
Get-NetFirewallRule -DisplayName "LAB-Block-ExampleCom" -ErrorAction SilentlyContinue
Remove-NetFirewallRule -DisplayName "LAB-Block-ExampleCom"

# — Enable Firewall Drop Logging —————
netsh advfirewall set allprofiles logging droppedconnections enable
netsh advfirewall show allprofiles | findstr -i "log"

# — Read Drop Log —————
Get-Content "$env:SystemRoot\system32\LogFiles\Firewall\pfirewall.log" | Select-Object -Last 30

# — Event Viewer — Firewall Changes —————
Get-WinEvent -LogName "Microsoft-Windows-Windows Firewall With Advanced Security/Firewall" `
  -MaxEvents 10 | Format-List TimeCreated, Id, Message

# — Network Protection —————
Get-MpPreference | Select-Object EnableNetworkProtection
Set-MpPreference -EnableNetworkProtection AuditMode # 2 = audit only
Set-MpPreference -EnableNetworkProtection Enabled # 1 = block
Set-MpPreference -EnableNetworkProtection Disabled # 0 = off

# — Enable Audit Logging for WFP Blocked Connections —————
auditpol /set /subcategory:"Filtering Platform Connection" /failure:enable
```

Key Event IDs

Event ID	Description
2004	Firewall rule added
2006	Firewall rule deleted
2082	Firewall setting changed (logging, profile settings)
5152	WFP blocked a packet (requires audit policy enabled)
5157	WFP blocked a connection
1125	Network Protection blocked URL — Audit Mode
1126	Network Protection blocked URL — Enforcement Mode

GUI Navigation Paths

Tool	Path
<code>wf.msc</code>	Win+R → wf.msc → Outbound Rules → New Rule
Firewall logs	wf.msc → top node → Properties → [Profile] tab → Logging → Customize
Windows Security NP	Windows Security → App & browser control → Reputation-based protection settings
Event Viewer	eventvwr.msc → Apps & Services → Microsoft → Windows → Windows Firewall → Firewall
Audit Policy	secpol.msc → Advanced Audit Policy → Object Access → Filtering Platform Connection